

GCR-4: A Philosophical and Geometric Treatise on Nonlinear State Projection

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Abstract

This treatise presents the philosophical, geometric, and perceptual foundations of the GCR-4 method, a nonlinear state-projection framework designed to preserve curvature, trajectory, and dynamical continuity. Unlike PCA [4], t-SNE [5], or UMAP [6], GCR-4 employs decreasing interval scaling, autoscaling to the $[0, 1]$ domain, and multi-view geometric capacity in the CSpace 1.01.26 software. These features position GCR-4 as a cutting-edge technology that constructs a spatial manifold rather than a fixed projection. The method is interpreted through natural perceptual laws [1], adaptive numerical strategies [10], and geometric ontology [9], establishing GCR-4 as a fundamentally new category of data representation.

1 Introduction

Classical dimensionality-reduction methods—PCA [4], t-SNE [5], and UMAP [6]—operate under the assumption that data consist of points in a metric space. Their goal is to produce a single two-dimensional projection that preserves certain statistical or topological properties. These methods do not construct geometric objects; they construct images.

Geometric Coordinates Reduction to 4 (GCR-4) method represents a departure from this paradigm. It treats data as *states* rather than points, emphasizing curvature, trajectory, and dynamical continuity. Its nonlinear scaling, autoscaling to $[0, 1]$, and multi-view geometric capacity position it as a cutting-edge technology [14] that transcends the limitations of classical projection methods.

2 Ontology of States

The philosophical foundation of GCR-4 lies in the interpretation of data as geometric states. A state is not merely a coordinate; it is a position on a trajectory, a point on a curve, a moment in a dynamical process. This ontological shift has profound implications:

- states possess direction,
- states possess curvature,
- states exhibit continuity,
- states belong to trajectories.

GCR-4 preserves these properties by constructing a manifold that reflects the geometry of state transitions, consistent with the principles of differential geometry [9].

3 Decreasing Intervals as Natural Adaptive Geometry

A defining feature of GCR-4 is its use of decreasing intervals along the projection axis. This nonlinear scaling is not arbitrary; it reflects natural mechanisms of perception and adaptive resolution.

3.1 Natural Connotations

Decreasing intervals correspond to phenomena observed in nature:

- **Logarithmic scales** (dB, pH, stellar magnitude) compress large values and expand small ones [3].
- **Psychophysical perception** follows the Weber–Fechner law [1], where sensitivity is highest at low stimulus intensities.
- **Biological sensing** emphasizes relative rather than absolute differences [2].

Thus, decreasing intervals in GCR-4 mirror natural perceptual systems that allocate more resolution to meaningful changes.

3.2 Technological Connotations

In technology and numerical analysis:

- adaptive meshes increase resolution in regions of high curvature [10],
- dynamic time warping expands segments with rapid change [11],
- compression algorithms allocate bits where detail is greatest.

GCR-4 adopts the same principle: stable regions are compressed, dynamic regions expanded.

4 Autoscaling to 0–1 as Natural Geometric Perception

Autoscaling to the interval $[0, 1]$ is central to GCR-4. It creates a domain of maximal geometric sensitivity.

4.1 Maximal Sensitivity

Mapping data to $[0, 1]$ ensures:

- proportional gradients,
- stable curvature,
- comparable distances across dimensions.

This makes $[0, 1]$ a natural space for detecting differences in state.

4.2 Psychophysical Foundation

The Weber–Fechner law [1] states that perceptual sensitivity decreases with stimulus intensity. Autoscaling to $[0, 1]$ reproduces this behavior:

- small differences are amplified,
- large differences are compressed,
- perception becomes nonlinear.

GCR-4 therefore aligns with biological mechanisms of perception.

4.3 Adaptive Numerical Foundation

Autoscaling also mirrors adaptive numerical strategies:

- stable regions are compressed,
- dynamic regions are expanded,
- resolution increases where curvature is high [10].

This makes GCR-4 an adaptive geometric detector of state transitions.

5 Cutting-Edge Technological Nature of GCR-4

GCR-4 qualifies as cutting-edge technology because it introduces capabilities absent from classical methods:

- nonlinear adaptive scaling,
- geometric state preservation,
- multi-view manifold construction,
- perceptual alignment with natural systems.

It does not refine existing methods; it defines a new category.

6 Multi-View Geometry

A unique property of GCR-4 is its ability to generate a spatial manifold in the CSpace 1.01.26 software from which multiple two-dimensional views can be extracted.

6.1 Spatial Manifold Construction

GCR-4 constructs a geometric object $M \subset \mathbb{R}^n$ that preserves:

- curvature,
- trajectory,
- dynamical continuity.

This object can be rotated, reprojected, and examined from multiple angles.

6.2 Contrast with Classical Methods

PCA [4], t-SNE [5], and UMAP [6] produce fixed 2D images:

- PCA yields a single linear projection,
- t-SNE yields a single nonlinear embedding,
- UMAP yields a single graph-based layout.

None produce a manifold that supports multi-view geometry.

6.3 Geometric Interpretation

Any 2D view corresponds to a projection

$$\pi : M \rightarrow \mathbb{R}^2,$$

chosen to emphasize curvature, transitions, or stability. The manifold remains constant; only the viewpoint changes.

7 Conclusion

GCR-4 integrates nonlinear scaling, autoscaling to $[0, 1]$, natural perceptual laws [1], adaptive numerical strategies [10], and multi-view geometric capacity. These features establish GCR-4 as a fundamentally new approach to state representation, transcending the limitations of PCA, t-SNE, and UMAP. Its philosophical foundation lies in the ontology of states, its geometric foundation in curvature and trajectory, and its technological foundation in adaptive nonlinear perception.

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